

Parent's Signature:

CANDIDATE'S NAME: _____ CTG: _____

YISHUN JUNIOR COLLEGE
2011 JC2 PRELIMINARY EXAMINATION

CHEMISTRY
HIGHER 2

9647/02

Paper 2 Structured Questions

MONDAY 22 AUGUST 2011
1400hrs – 1600hrs
(2 hours)

Candidates answer on the Question Paper

Additional Materials: Data Booklet

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READ THESE INSTRUCTIONS FIRST

Write your name and CTG on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

Paper 1

Total / 40

Paper 2

1 / 12

2 / 14

3 / 20

4 / 15

5 / 11

Total / 72

Paper 3

Total / 80

Overall / 192

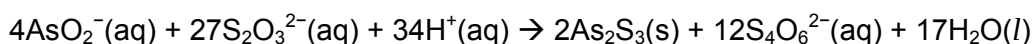
1 Planning (P)

One method of studying the kinetics of a reaction is using the initial rate method. To determine the initial rate we can time how long it takes to reach an identifiable point early in the reaction.

When aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, is added to a solution containing aqueous sodium metaarsenite, NaAsO_2 , that is acidified with ethanoic acid, CH_3COOH , a bright yellow suspension of solid arsenic(III) sulfide, As_2S_3 , will be formed after some time. This is the identifiable point in the reaction.

It is believed that the rapid precipitation of a large amount of bright yellow solid after a short time interval is due to the build-up of the concentration of arsenic(III) sulfide to that of a supersaturated solution before precipitation begins.

The equation for the overall reaction can be taken to be:



A student carried out a series of preliminary experiments, using approximate volumes of the three reagents, and each solution was made up to the same total volume with distilled water. He found that the time taken for the bright yellow precipitate to appear doubled when the volume of sodium thiosulfate added was halved.

- (a) Briefly explain why it is necessary to top up the solution with distilled water.

[1]

- (b) Assuming that the rapid precipitation occurred when the same amount of arsenic(III) sulfide was produced in all experiments, state the simple relationship between the time taken for the bright yellow precipitate to appear and the initial rate of reaction.

[1]

- (c) (i) Using the results of the preliminary experiments, predict the relationship between the concentration of sodium thiosulfate and the rate of reaction.

-
- (ii) State what can be deduced regarding the role of the thiosulfate ion, $\text{S}_2\text{O}_3^{2-}$, in the mechanism of the reaction, based on your prediction in (c)(i).

[2]

- Experiment 1 is carried out using the quantities given in the table below:

Experiment	Volume of 2.00 mol dm ⁻³ Na ₂ S ₂ O ₃ / cm ³	Volume of 0.020 mol dm ⁻³ NaAsO ₂ / cm ³	Volume of 2.00 mol dm ⁻³ CH ₃ COOH / cm ³	Volume of water / cm ³	time taken for the bright yellow ppt to appear / s
1	20	20	40	20	t ₁
2					t ₂

- Your plan should include:

- o the apparatus used to measure the various volumes;
- o the sequence that the various solutions are mixed;
- o how the time for the appearance of the bright yellow precipitate is measured; and
- o other experimental details to ensure the consistency of the experiment.

[illegible]

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- (e) Taking the time for the bright yellow precipitate to appear in Experiments 1 and 2 to be t_1 and t_2 respectively, explain how the results of the two experiments can be used to determine the order of reaction with respect to sodium metaarsenite.

[2]

- (f) Aqueous sodium metaarsenite and solid arsenic(III) sulfide were identified to be potential safety hazards in this experiment. Explain briefly how they are hazardous, and state how you would minimise the risk.

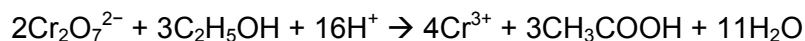
[1]

[Total: 12 marks]

- 2 Chromium occurs in our environment primarily in two valence states, chromium(III), and chromium(VI). Long term exposure to chromium is known to cause damage to liver and nerve tissues, and studies have clearly established chromium(VI) to be a human carcinogen.

- (a) Potassium dichromate(VI), $\text{K}_2\text{Cr}_2\text{O}_7$, is a strong oxidising agent commonly used in organic chemistry to determine the concentration of ethanol, $\text{C}_2\text{H}_5\text{OH}$.

Potassium dichromate(VI) and ethanol reacts according to the following equation:



In a typical determination, a 10.0 cm^3 sample of an aqueous solution of ethanol was completely reacted with 50.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ potassium dichromate solution in acidic medium. The excess dichromate(VI) in 15.0 cm^3 of the resultant solution required 22.50 cm^3 of 1.00 mol dm^{-3} of iron(II) sulfate, FeSO_4 , for complete reaction.

- (i) Calculate the amount of dichromate(VI) ions present in 15.0 cm^3 of the resultant solution, given that one mole of $\text{Cr}_2\text{O}_7^{2-}$ required six moles of Fe^{2+} for complete reaction.

- (ii) Calculate the amount of dichromate(VI) ions that reacted with the ethanol solution.

- (iii) Calculate the concentration of ethanol in the 10.0 cm^3 sample.

[4]

- (b) Chromium plating is an important industrial application of electrolysis, where a steel object is plated with a thin layer of chromium to make it resistant to rusting and scratches.

In a typical process, one electrode is usually made of graphite while the other is the steel object to be electroplated, and the electrolyte used is chromium(III) sulphate, $\text{Cr}_2(\text{SO}_4)_3$.

- (i) State whether the steel object should be connected to the positive or negative pole of the battery.

- (ii) Calculate the maximum mass of chromium that will be plated onto the steel object if a 10 A current is passed through the cell for 30 minutes.

- (iii) In this method of electroplating, the graphite electrode has to be replaced from time to time. Suggest a reason for this.

[4]

- (c) Sparingly soluble chromium(III) compounds like chromium(III) hydroxide are not considered health hazards, as only limited amount will be absorbed by cells.

A student investigated the solubility of chromium(III) hydroxide by adding solid sodium hydroxide to a 100 cm³ sample of chromium(III) chloride.

- (i) The solubility of chromium(III) hydroxide in water is $1.27 \times 10^{-6} \text{ g dm}^{-3}$ at 25 °C. Calculate the value of the solubility product, K_{sp} , of chromium(III) hydroxide, stating the units.

- (ii) Given that the concentration of the chromium(III) ions in the 100 cm³ sample is $1.0 \times 10^{-5} \text{ mol dm}^{-3}$, calculate the minimum mass of sodium hydroxide added that will cause chromium(III) hydroxide to be precipitated.

- (iii) When an excess of sodium hydroxide was added to the precipitate of chromium(III) hydroxide, it dissolves. Explain why this is so.

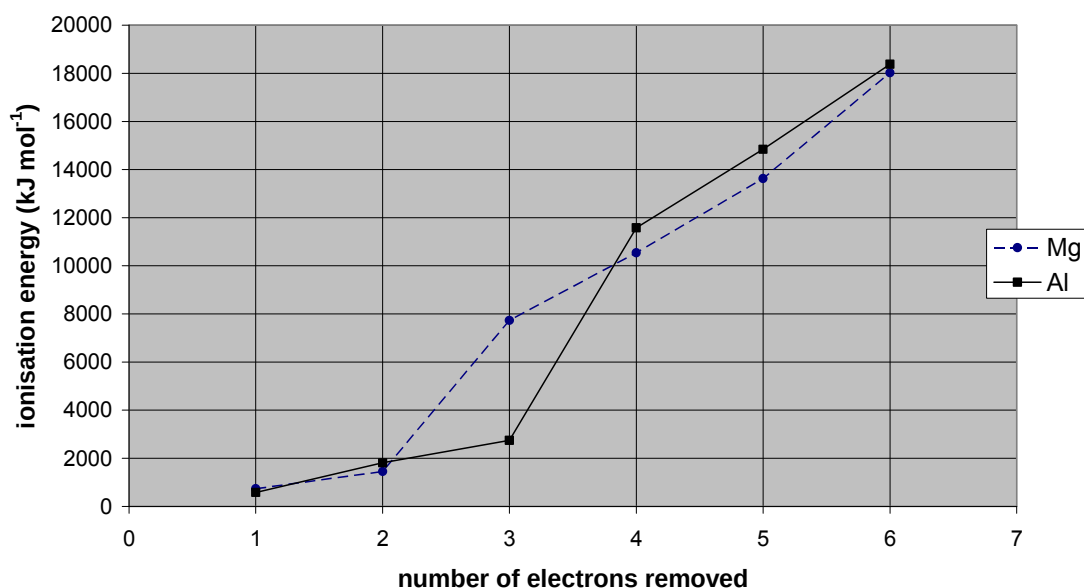
[6]

[Total: 14 marks]

- 3 Magnesium, like aluminium, is a fairly strong and light-weight metal that is commonly used to make body panels of racing cars.

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- (a) The graph of successive ionisation energy of magnesium and aluminium are given below:



- (i) Write an equation to represent the second ionisation energy of magnesium.

- (ii) Explain why the ionisation energies increase for both magnesium and aluminium as electrons are successively removed.

- (iii) Explain why the third ionisation energy of magnesium is much higher than that of aluminium.

[4]

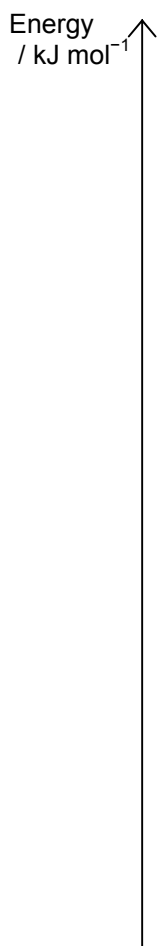
- (b) Magnesium nitride, Mg_3N_2 , is produced when magnesium metal is burnt in pure nitrogen atmosphere.

The lattice energy of magnesium nitride can be calculated from a Born-Haber cycle using the following data:

	$\Delta H / \text{kJ mol}^{-1}$
standard enthalpy change of atomisation of magnesium	+148
sum of the first three electron affinities of nitrogen	+146
standard enthalpy change of formation of magnesium nitride	-461

- (i) Define what is meant by the term *lattice energy* of an ionic compound.

- (ii) Using relevant data from the *Data Booklet* and the information given above, construct the Born-Haber cycle for magnesium nitride in the axis below, and use the cycle to calculate the lattice energy of magnesium nitride.



- (iii) The standard enthalpy change of formation of magnesium oxide is -601 kJ mol^{-1} . With reference to the energy changes in the Born-Haber cycle, suggest why the formation of magnesium oxide is more exothermic than magnesium nitride.

[7]

- (c) Describe the reactions of the Group II metals, magnesium to barium, with water and write a general equation for the reaction.

[3]

- (d) Magnesium nitrate undergoes thermal decomposition at a much lower temperature (330°C) than barium nitrate does (590°C).

- (i) Write an equation, including state symbols, for the decomposition of magnesium nitrate.

- (ii) Suggest an explanation for the above observation.

- (iii) The decomposition reaction of magnesium nitrate is endothermic and the entropy change, ΔS , is positive.

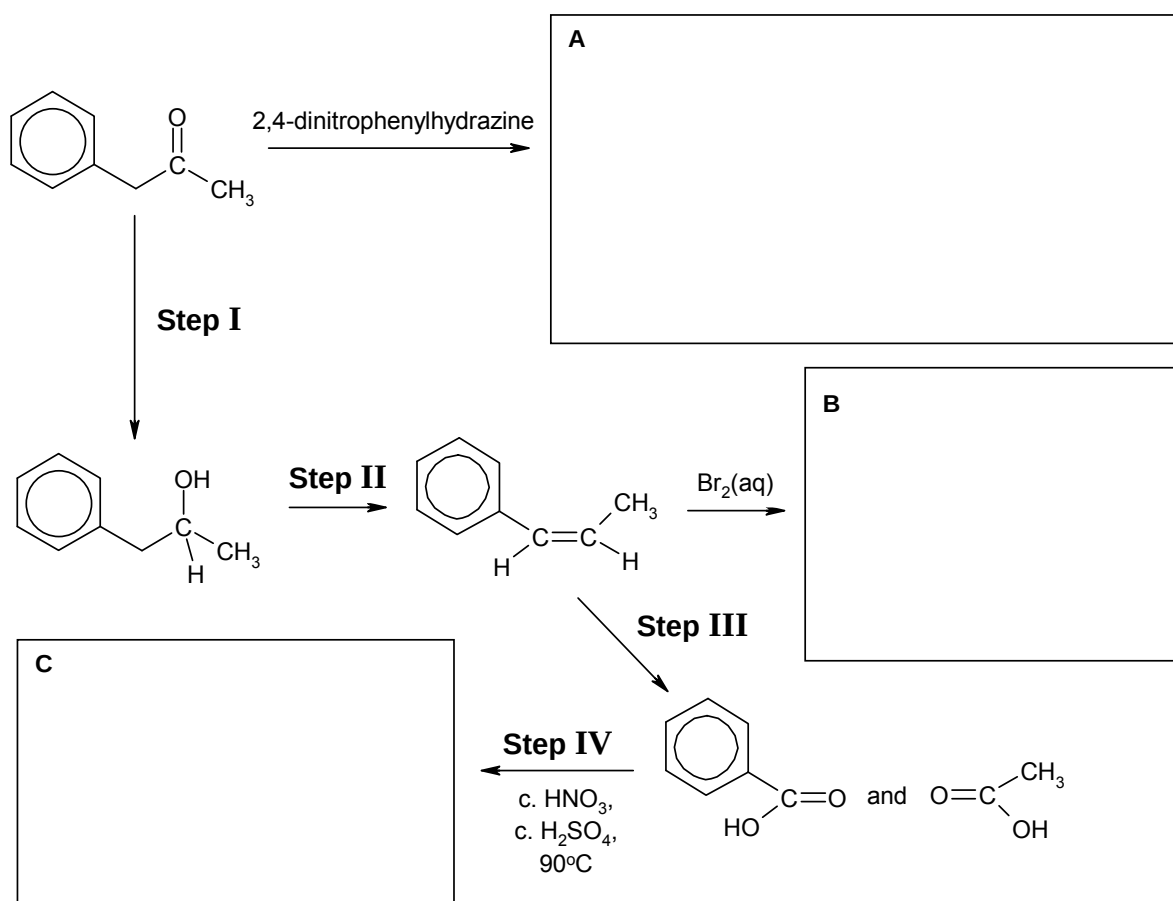
Explain briefly why ΔS is positive, and suggest how the feasibility of the reaction changes with increasing temperature.

[6]

[Total: 20 marks]

- 4 Phenylpropanone, $\text{C}_6\text{H}_5\text{CH}_2\text{COCH}_3$, is used in the manufacture of amphetamine, a stimulant commonly used to treat Attention Deficit Hyperactivity Disorder (ADHD). Due to the illicit use in clandestine chemistry, it was made a controlled substance in 1979 in the United States.

A reaction scheme beginning with phenylpropanone is shown below.



- (a) (i) Draw the structures of compounds **A**, **B** and **C** in the boxes above.

- (ii) Suggest reagents and conditions for steps **I** to **III** by completing the table below.

Steps	Reagents and conditions
I	
II	
III	

[6]

- (b) (i) State the type of reaction for step **II**.

- (ii) The **major** product of step **II**, 1-phenylpropene exists as a pair of stereoisomers.

A. Draw the displayed structural formulae of the two isomers of 1-phenylpropene, to illustrate the type of stereoisomerism involved.

B. Give the IUPAC name of the **minor** product of the reaction.

[3]

- (c) (i) Outline the mechanism of the reaction in step **IV**, showing clearly the structure of the reaction intermediate, and using curly arrows to show the movement of electrons.

- (ii) With reference to the rate determining step of the mechanism, briefly explain how concentrated sulfuric acid is able to speed up this reaction.

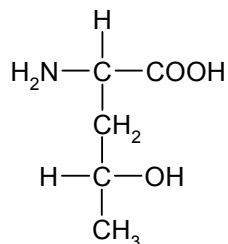
- (iii) Given that the range of temperature used for the nitration of benzene is usually between 55 °C and 60 °C, give a reason why the reaction in step **IV** requires a higher temperature.

[6]

[Total: 15 marks]

- 5 Natural polypeptides and proteins are made up of 20 different α -amino acids, differing in the nature of the R group.

(a) The structure of threonine, an α -amino acid, is given below.



- (i) Explain what is meant by an α -amino acid.

- (ii) Threonine exhibits stereoisomerism.

State the type of stereoisomerism and draw two structures to illustrate this isomerism.

Type of isomerism: _____

Structures:

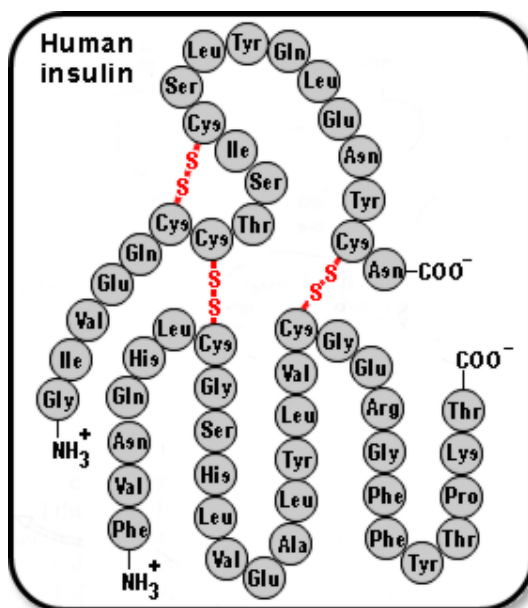
- (iii) Give the structural formulae of the organic products formed when threonine is reacted with sodium carbonate, and heated with bromoethane in ethanol respectively.

with sodium carbonate	with bromoethane in ethanol

[5]

- (b) Insulin is a hormone central to regulating carbohydrate and fat metabolism in the body. Insulin causes cells in the liver, muscle, and fat tissue to take up glucose from the blood, storing it as glycogen in the liver and muscle.

The sequence of human insulin, shown below, is made up of two chains joined by three covalent -S-S- bonds.



- (i) Human insulin is denatured when heavy metal ions are present in the body. Explain the term denaturation and explain how denaturation of insulin occurs due to the action of heavy metal ions.

- (ii) The secondary structure of insulin consists of segments of α -helices. With the aid of a diagram, illustrate how the polypeptide chain is held in the structure of an α -helix.

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